

ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

88yGyDdvVRHcVsZF6obPaV3Qov7oX4L1HzAA15uboop

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Compounds: Xanax, Adderall, Cocaine

RESEARCH PURPOSES ONLY

This analysis is generated for research and educational purposes. The predictions and insights provided should not be considered medical advice.

New Compound Analysis

Name: Adcocanax

Formula: C₂₁H₂₈N₄O₄

Weight: 456.48

Drug-likeness: 0.45

Activity:

Stimulant with anxiolytic properties

Target Analysis

D1 dopamine receptor

Binding Affinity: -8.5

D1 receptors are implicated in modulating mood and are targeted for depression and attention disorders.

GABA-A receptor

Binding Affinity: -9.2

GABA-A receptors are critical in inhibiting neural overactivity, relevant for anxiety management.

Toxicity Profile

Risk Level: Moderate to high

Affected Systems: Heart, Brain, Liver

Mitigation Suggestions:

- Dose regulation
- Combination with protective agents
- Regular monitoring of cardiac function

Pharmacokinetics

Absorption:

Rapid absorption via oral route

Distribution:

High distribution with significant plasma protein binding

Metabolism:

Extensively metabolized by liver enzymes CYP3A4/5

Excretion:

Primarily renal excretion with a small portion through feces

Half-life: Approximately 8 hours

Detailed Analysis

Adcocanax is a synthesized compound that amalgamates pharmacodynamic qualities drawn from Adderall, Cocaine, and Xanax. This compound leverages the dopamine transporter and GABA-A receptor modulation, creating a balancing mechanism in treating disorders characterized by both deficient attention and heightened anxiety. In silico analyses indicate potential high binding affinity to selected neuroreceptors, while the anticipated drug-likeness score suggests moderate alignment with Lipinski's rule of five, subject to further pharmacological refinements. Nevertheless, its profile carries inherent risks of neuro-cardiac stress and dependency, recommending cautious clinical exploration yet highlighting possibilities for addressing comorbid neuropsychiatric conditions.

Combine base compounds, simulate new drug possibilities with AI, and earn rewards through Solana blockchain. Join our community of researchers and innovators in shaping the future of pharmaceutical discovery.

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